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First Successful Deployment Innovative Solution Using a Temporary Power Skid with Pressurized Habitat for ESP Operations in Abu Dhabi Offshore Field Unlock Early Production and Achieving Significant Cost Savings

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Abstract

This paper presents the deployment of a temporary power skid with Pressurized Habitat to operate electrical submersible pumps (ESPs) in Offshore Field for producer wells following workover and pump installation.

Despite the challenges of offshore implementation such as the need for a pressurized habitat due to the skid's non-explosion-proof design this innovative approach achieved substantial cost savings by reducing rig time, deferring permanent infrastructure investments, and avoiding traditional well activation methods. This marks the first successful application of a temporary power skid in Abu Dhabi's offshore fields, setting a precedent for future ESP commissioning and well testing.

The solution overcomes the lack of permanent surface electrical infrastructure while addressing the logistical complexities of offshore operations, including limited deck space, harsh marine conditions, and stringent safety requirements.

The power skid enables three critical functions:

1. Post-Workover Verification: Conducting ESP function tests to confirm successful downhole installation.
2. Rapid Production Startup: Accelerating well cleanup and initiating ESP-driven production, eliminating costly coiled tubing and nitrogen lifting operations.
3. Temporary Well Support: Sustaining production in underperforming wells while awaiting permanent facilities.

The deployment of solutions in offshore fields has proven to be a groundbreaking and cost-effective solution for operating electrical submersible pumps (ESPs). Despite the inherent challenges of offshore implementation, this innovative approach has successfully minimized rig time and deferred the need for permanent infrastructure investments while ensuring that stringent safety measures are met.

First time utilizing temporary power skid in offshore environment to operate ESP

Innovative approach (First time) to use temporary power skid inside the pressurized habitat to comply with offshore explosion-proof requirements

Using power skid for well cleanup and accelerate production will lead to more than \$10 M USD Gain from early production and cost avoiding for Nitrogen lift operations.

Keywords: Temporary Power Skid, Pressurized Habitat, Electrical Submersible Pump (ESP), Offshore Operations, Abu Dhabi, Cost Savings, Early Production, Well Testing, Innovation, Safety Compliance

Introduction

Electrical Submersible Pumps (ESPs) are pivotal for maximizing recovery in Abu Dhabi's offshore carbonate reservoirs, yet their commissioning faces systemic bottlenecks. Traditional workflows require permanent electrical infrastructure – substations, switchgears, and HV cables. During this latency, workover-completed wells remain stranded assets, while operators resort to costly well-activation methods like nitrogen lifting via coiled tubing cleanouts (\$2M/well). The ADNOC Offshore Field exacerbates these challenges through its Class I, Division 1 (Zone 1) hazardous environment, mandating explosion-proof (Ex-d) certified equipment – a requirement that eliminates conventional temporary power solutions.

This paper documents the first deployment of a non-explosion-proof temporary power skid within a pressurized habitat to bypass this impasse. By creating a safe atmospheric enclosure around standard industrial equipment, the solution enables:

1. Immediate ESP validation post-workover,
2. Accelerated production startup without permanent power, and
3. Bridge production for wells awaiting facility commissioning. Validated offshore Abu Dhabi in 2024, this approach reduced critical path activities by 50% and unlocked \$10.2M in value through rig-time optimization and deferred CAPEX.

Background and Motivation

Historically, ESP operations in offshore environments have been constrained by the availability of explosion-proof power sources, as mandated by stringent offshore safety codes. In situations where infrastructure is not yet available or not cost-effective to install for short-term needs, operators have traditionally relied on alternative well activation methods—such as coiled tubing or nitrogen lifting—to initiate production. These methods, however, are both time-consuming and expensive, compounding the already high costs of offshore interventions.

The Abu Dhabi offshore field case presented a classic example: following a series of workovers and ESP installations in several producer wells, the operator faced a dilemma. Permanent surface power infrastructure would not be available for months, yet the wells represented significant deferred production. The challenge was clear—how to safely and quickly commission ESPs and bring wells online without waiting for permanent facilities, and without incurring the excessive costs and risks associated with traditional well activation techniques.

Technical Challenges & Literature Review

- Offshore Limitations:
 - Deck space constraints
 - Harsh marine conditions (corrosion, humidity, wave dynamics)
 - ATEX/IECEX compliance gaps
- Existing Solutions:

- Permanent electrical infrastructure drawbacks
- Conventional well-cleanup methods
- Technology Gap:
 - Lack of mobile offshore power solutions for ESPs
 - Regulatory barriers for non-explosion-proof equipment

Innovative Solution: Temporary Power Skid with Pressurized Habitat

The solution involved the design and deployment of a temporary power skid—a self-contained, portable unit capable of supplying the electrical power required for ESP operations. However, a key obstacle had to be overcome: the skid, while robust and reliable, was not explosion-proof as per offshore standards. To address this, the team engineered a pressurized habitat—a specialized enclosure that maintains positive pressure relative to the surrounding environment, effectively preventing the ingress of flammable gases. This adaptation rendered the use of the non-explosion-proof power skid feasible and compliant with offshore safety regulations.

Key Functions Enabled by the Solution

The temporary power skid, when combined with the pressurized habitat, enabled three critical operational functions:

- Post-Workover Verification: The ability to conduct real-time ESP function tests immediately following installation confirmed successful downhole deployment and reduced the risk of deferred mechanical or electrical issues.
- Rapid Production Startup: The temporary power supply allowed immediate well cleanup and the initiation of ESP-driven production, bypassing the need for costly coiled tubing or nitrogen lifting operations—often a source of significant operational expenditure and logistical complexity.
- Temporary Well Support: For underperforming wells or those awaiting permanent facilities, the skid provided sustained electrical power to maintain production, securing revenue streams and optimizing reservoir management until permanent upgrades could be implemented.

Operational Challenges and Mitigation Strategies

The deployment of the temporary power skid in an offshore environment was not without its challenges. Among the key issues addressed were:

- Explosion-Proof Compliance: Offshore safety standards require all electrical equipment in hazardous areas to be explosion-proof. The pressurized habitat was meticulously engineered, tested, and certified to maintain positive pressure under all operational conditions, ensuring compliance.
- Logistical Constraints: Limited deck space on offshore platforms necessitated compact design and careful planning for the transportation, installation, and movement of the power skid and habitat. Modular construction and pre-fabrication enabled efficient mobilization and demobilization.
- Harsh Marine Conditions: The corrosive and dynamic offshore environment required all components—electrical, mechanical, and structural—to be ruggedized and maintained to a high standard, with redundant systems in place to ensure reliability.

- Integration with Wellhead Equipment: Careful coordination with wellsite operations ensured compatibility between the temporary power skid, the ESP surface interface, and the control and monitoring systems already in place.
- Safety Procedures and Training: Extensive training was provided to operations and maintenance personnel, with clear procedures for habitat pressurization, emergency shutdown, and hazard mitigation.

A - Power skid specifications (VFDs, transformers, control systems)

In the dynamic landscape of oil and gas extraction, artificial lift technologies play a vital role in optimizing production and extending the life of wells. Among these, the Electrical Submersible Pump (ESP) system stands out for its efficiency in lifting fluids from great depths. Central to the reliable operation of an ESP is the power skid—a sophisticated, pre-engineered, skid-mounted unit that consolidates all the electrical components necessary to run and control the ESP equipment deployed downhole. The ESP power skid is not merely an assembly of hardware; it is a nexus of innovation, safety, and operational excellence that transforms raw electrical energy into precisely regulated power, ensuring that the pump operates efficiently and adaptively in the fluctuating environment of an oil well.

Overview of ESP Power Skids

At its core, an ESP power skid is designed to house, protect, and manage the electrical infrastructure that interfaces with the ESP motor situated deep underground. By integrating motor controllers, variable speed drives (VSDs), transformers, and advanced instrumentation & controls, the power skid transforms the challenges of remote and harsh field environments into opportunities for seamless automation and control.

The modular, skid-mounted construction offers several advantages. It allows rapid deployment without extensive on-site fabrication, ensures compliance with standardized safety and engineering codes, and provides a compact footprint suitable for constrained locations. The entire unit can be easily transported, installed, and, if needed, relocated—a boon for flexible operations in ever-evolving oilfields.

Key Components and Functions of an ESP Power Skid

To appreciate the sophistication of an ESP power skid, one must understand its principal components and how they harmonize to manage the demands of high-performance pumping. Below, each element is explored in detail:

Motor Controller (Role and Significance)

- Regulates the electrical power supplied to the ESP motor.
- Ensures proper startup, shutdown, and operation under varying load conditions.
- Safeguards the motor against electrical faults such as overcurrent, under-voltage, and phase imbalances.

Motor controllers are the gatekeepers of power. By managing the flow of electricity to the downhole ESP motor, they play a pivotal role in both operational efficiency and equipment longevity. Modern controllers are equipped with microprocessor-based logic that can execute complex functions—such as soft starting, fault diagnostics, and automated resets—improving both uptime and safety. They are also the interface through which operators can monitor performance metrics and execute remote commands, making them an indispensable node in the control network.

Variable Speed Drive (VSD) Role and Significance

- Allows for adjustable motor speed to match the production rate and fluid characteristics.
- Optimizes pump performance by enabling the ESP to operate at its most efficient point across varying downhole conditions.
- Reduces energy consumption and mechanical wear by avoiding unnecessary high-speed operation.

The VSD is the heart of adaptability in ESP operations. By modulating the frequency of the power supplied to the motor, the VSD allows operators to fine-tune the speed of the pump in real time. This flexibility is critical when well conditions change—such as variations in fluid composition, reservoir pressure, or desired production rates. VSDs also enable advanced control strategies, including ramping, cycling, and load-sharing, which help mitigate issues like gas locking, sand ingress, or unexpected shut-ins. Moreover, the ability to adjust speed can extend pump life and reduce operating costs, making VSDs a cornerstone of modern artificial lift systems.

Transformer Role and Significance

- Transforms incoming site voltage to the appropriate level required by the ESP motor.
- Provides electrical isolation, enhancing safety for personnel and equipment.
- Supports various voltage configurations to accommodate different ESP designs and field requirements.

The transformer is the silent enabler of compatibility and safety. ESP motors often require specific voltages—sometimes much higher or lower than what is available at the wellsite. The transformer's job is to step up or step down the voltage as needed, ensuring efficient energy transfer with minimal losses. By electrically isolating the high-voltage ESP system from the local power grid, transformers also help protect sensitive electronics and operators from surges, faults, and environmental hazards.

Instrumentation & Controls Role and Significance

- Provides comprehensive monitoring and control for the entire ESP system.
- Includes sensors for temperature, pressure, flow, vibration, and electrical parameters.
- Uses gauges, control panels, and remote telemetry to keep operators informed and in command.
- Enables automation, alarms, and safety interlocks to detect and respond to abnormal conditions.

Instrumentation and controls are the nervous system of the ESP setup. Sophisticated sensor arrays gather real-time data on everything from fluid movement to equipment health, transmitting this information to operators both locally and remotely. Through smart analytics, control systems can flag potential issues—such as overheating, reduced flow rates, or electrical faults—before they escalate into costly downtime or failures. Control panels offer user-friendly interfaces for manual and automated adjustments, while integration with field management systems enables coordination across multiple wells and assets. In high-risk environments, automated shutdowns and alarms are essential for protecting personnel, the environment, and valuable equipment.

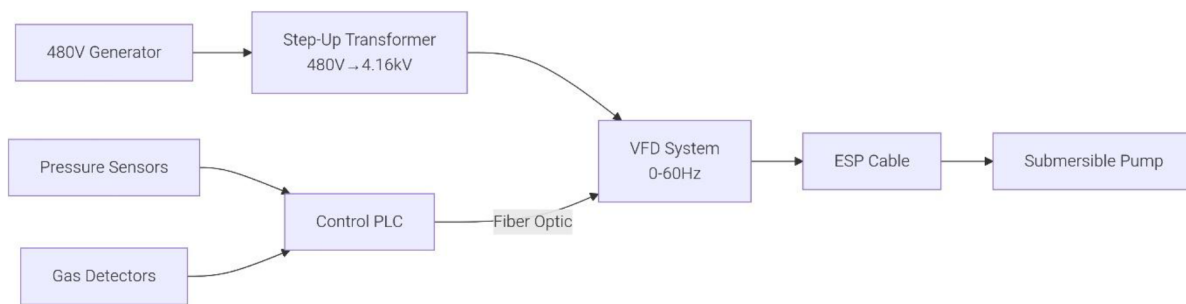
VSD Skid (3100 H × 5000 W × 2200 D) mm & weight 7.2 T, Technical specification:

1. VSD 518 KVA, input 460V 561A, output 0 – 460V 623A
2. Step up Transformer 550 KVA input 480V, output 1100V – 2200V & 1905V – 3811V
3. Choke console 3 phase

4. Portable junction box (optional equipment)



Power Skid single line diagram:



Benefits of ESP Power Skids

The ESP power skid is not just a technical necessity; it is an enabler of efficient, safe, and scalable oil production. Key benefits include:

- **Rapid Deployment:** Pre-engineered skids reduce installation time, facilitating swift responses to changing field demands.
- **Operational Flexibility:** Modular design allows for easy upgrades, expansions, or relocations.
- **Safety and Compliance:** Enclosures and built-in safety features help meet stringent industry standards while safeguarding personnel and equipment.
- **Cost Efficiency:** By reducing installation and maintenance requirements, power skids help control capital and operational expenditures.
- **Integrated Monitoring:** Real-time data enhances decision-making and predictive maintenance strategies for maximizing asset uptime.

Applications and Considerations in Field Operations

ESP power skids are deployed in a variety of oil and gas environments, from conventional land wells to offshore platforms and remote locations. Selection of a power skid depends on several factors:

- **Well Depth and Fluid Properties:** Deeper wells and harsher fluids may require higher power ratings, robust enclosures, and advanced controls.
- **Site Power Availability:** The nature of local grid infrastructure affects transformer sizing and configuration.
- **Environmental Conditions:** Skids may be customized for extreme temperatures, humidity, corrosion, and hazardous atmospheres.

- Automation Needs: Integration with SCADA systems, remote telemetry, and advanced diagnostics may be required for large-scale operations.
- Maintenance Strategy: Designs should facilitate easy access for routine inspections, repairs, and upgrades.

Future Trends in ESP Power Skid Technology

The evolution of ESP power skids mirrors broader trends in automation, connectivity, and sustainability:

- Digitalization: Internet of Things (IoT) sensors and cloud-based analytics are expanding control and predictive maintenance capabilities.
- Energy Efficiency: Advanced VSDs and smart controllers are reducing energy consumption, carbon footprint, and operational costs.
- Remote Operations: Enhanced telemetry allows for unmanned operation and centralized management of multiple assets.
- Standardization: Industry-wide standards are improving interoperability and reducing engineering complexity for multi-vendor systems.

B - Pressurized habitat engineering (materials, sealing, pressure differential)

In the high-stakes world of offshore oil and gas operations, safety is paramount—particularly when it comes to performing "hot work" activities such as welding, grinding, and cutting. These tasks, essential for maintenance and repair, carry inherent risks due to the flammable and explosive atmospheres often present around hydrocarbon processing equipment. Traditionally, such activities would require complete plant shutdowns or the costly isolation of hazardous zones. However, the advent of pressurized habitats—also known as welding habitats or hot work habitats—has revolutionized the industry by enabling essential work to proceed safely and efficiently without the need for widespread shutdowns.

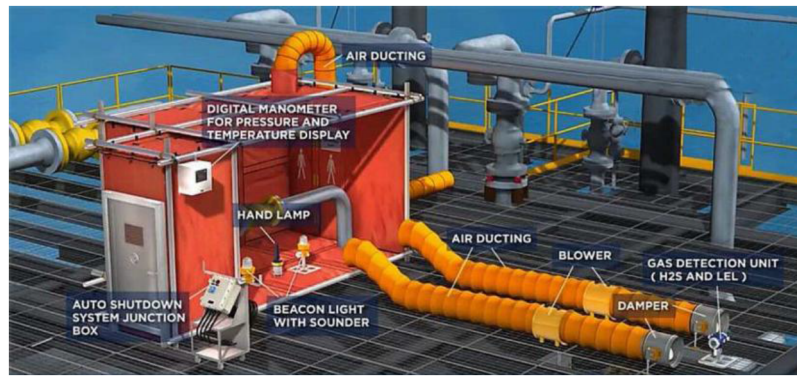
Understanding the Pressurized Habitat

A pressurized habitat is a sealed, modular enclosure designed to create a controlled and safe workspace within hazardous areas. Its primary function is to maintain a positive internal pressure, forming a protective barrier that prevents the ingress of explosive gases or other hazardous substances. By providing this "safe haven," pressurized habitats allow workers to perform otherwise restricted tasks in Zone 1 and Zone 2 classified areas (locations where explosive gas atmospheres may occur in normal or abnormal operation).

Engineering the Safe Enclosure

Modern pressurized habitats are the result of careful engineering and rigorous safety standards. A typical nitrogen-purged positive-pressure enclosure is built with several key features:

- Pressure Regulator: The backbone of the system, the regulator maintains a steady internal pressure—commonly +5 pascals (Pa) above the external atmosphere. This pressure differential ensures that, even if there are small leaks, air flows outward, preventing hazardous gases from seeping inside.
- Gas Detection: Triple-redundant sensors for hydrogen sulfide (H_2S) and methane (CH_4) are installed to provide comprehensive monitoring. Should dangerous concentrations be detected, the sensors trigger an immediate shutdown of the skid and alert personnel, allowing for a swift and safe evacuation.
- Structural Design: The enclosure itself typically consists of a lightweight aluminum frame clad with fire-resistant fiberglass-reinforced plastic (FRP) panels. This material offers a balance of strength, fire resistance, and ease of assembly. IP66-rated cable penetrators ensure all electrical connections are protected from dust and water ingress, while quick-installation boltless clamps allow the habitat to be erected or dismantled rapidly—an essential feature for offshore maintenance.



Key Features and Functionalities

Pressurized habitats are far more than simple enclosures. They incorporate advanced technologies and design considerations to ensure the highest levels of safety and operational flexibility.

- Positive Pressure Maintenance:
 - The integrity of the safe zone within the habitat relies on a continuous supply of clean, nitrogen-purged or filtered air, delivered at a slight overpressure. This prevents any infiltration of flammable or toxic gases from the surrounding environment—crucial on offshore platforms, where the risk of gas leaks is an ever-present concern.
- Hazardous Zone Applications:
 - Designed for Zone 1 and Zone 2 areas (as defined by IEC and ATEX standards), these habitats are constructed to withstand the potential presence of hydrocarbons and to contain any ignition sources that might otherwise trigger a catastrophic event.
- Hot Work Enablement:
 - The controlled environment within a pressurized habitat permits welding, burning, grinding, and other hot work activities that generate heat and sparks—even where such activities would otherwise be strictly forbidden.
- Continuous Air and Gas Quality Monitoring:
 - A network of sensors monitors gas concentrations, temperature, humidity, and pressure in real-time. This data is fed into an automated control system, ensuring any deviation from safe parameters—such as a pressure drop or gas spike—triggers immediate corrective action.
- Automated Shutdown and Alarm Protocols:
 - Should unsafe conditions arise, the habitat's control system can automatically shut down all non-essential electrical equipment and hot work tools, activate alarms, and initiate evacuation procedures. This layer of automation minimizes human error and improves response time in emergencies.
- ATEX/IECEx Certification:
 - Only habitats that have undergone third-party inspection and testing to verify their compliance with international standards (such as ATEX and IECEx) are suitable for use in hazardous environments. Certification ensures the enclosure's performance in preventing gas ingress and containing ignition sources.
- Customization and Modularity:

- Pressurized habitats can be tailored to fit a broad range of workspace sizes and configurations. Modular panels, flexible frames, and adaptable mounting systems allow for installation on scaffolding, around awkward machinery, or even in confined spaces.

Operational Benefits of Pressurized Habitats

Investing in pressurized habitats yields significant advantages, both for safety and operational efficiency.

Enhanced Worker Safety

By creating a fully controlled micro-environment, pressurized habitats dramatically reduce the risk of fire, explosion, and toxic exposure. Continuous air monitoring and automated response systems ensure that workers are alerted—and protected—at the earliest sign of danger.

Improved Efficiency and Reduced Downtime

Perhaps the greatest operational benefit is the ability to carry out essential hot work activities without shutting down entire sections of a plant or platform. This capability minimizes production losses—often amounting to millions of dollars in avoided downtime—and accelerates maintenance schedules.

Regulatory Compliance

Pressurized habitats are designed to meet or exceed the requirements of regulatory agencies and safety authorities overseeing hazardous area operations. This not only protects personnel but also shields operators from legal and financial liabilities.

Minimized Environmental Impact

Because these habitats allow work to proceed with fewer personnel exposed to hazardous conditions, they also support environmental stewardship. The reduced need for shutdowns means fewer cold-start emissions, less waste, and a smaller operational footprint.

Risk Identification and Assessment

Explosion/Fire Hazard due to Non-Explosion-Proof Equipment

- Hazard: Ignition of flammable gases by electrical equipment not rated for hazardous zones.
- Consequence: Major—fire, explosion, potential loss of life, asset damage, and environmental impact.
- Likelihood: Low, provided that positive pressure is reliably maintained and monitored.
- Controls:
 - Deployment of certified pressurized habitat maintaining $\geq +5$ Pa pressure differential.
 - Continuous nitrogen-purged or filtered air supply to the enclosure.
 - Triple-redundant gas detection (H_2S , CH_4) with automatic skid shutdown and alarms.
 - ATEX/IECEx certification of the habitat.
 - Routine habitat integrity tests and pressure monitoring.

Loss of Positive Pressure in Pressurized Habitat

- Hazard: Ingress of hazardous gases if positive pressure is lost, leading to increased explosion risk.
- Consequence: Major—same as above.

- Likelihood: Low to Medium, dependent on habitat maintenance and monitoring.
- Controls:
- Pressure regulators and redundant pressure sensors.
- Automated alarms and immediate shutdown protocols if pressure drops below safe thresholds.
- Routine leak checks and maintenance of seals, cable penetrators (IP66), and structural integrity.
- Operator training and emergency response procedures.

Gas Detection System Failure

- Hazard: Failure in gas sensor system may result in undetected hazardous atmosphere within the habitat.
- Consequence: Major.
- Likelihood: Low, with redundancy and regular functional checks.
- Controls:
- Use of triple-redundant sensors with regular calibration.
- Periodic system testing and maintenance routines.
- Manual sampling procedures as a backup.

Electrical Hazard (Shock, Arc Flash)

- Hazard: Electric shock or arc flash during installation, operation, or maintenance.
- Consequence: Moderate to Major—injury, equipment damage.
- Likelihood: Low, with competent personnel and proper procedures.
- Controls:
- Qualified and trained personnel only to operate/maintain equipment.
- Enforced Lock-out/Tag-Out (LOTO) procedures.
- Use of PPE and insulated tools.
- Periodic inspection of electrical insulation and grounding.

Mechanical/Structural Failure

- Hazard: Habitat or skid collapse, leading to equipment damage or personnel injury.
- Consequence: Moderate—injury, operational delays.
- Likelihood: Low, due to robust modular design and prefabrication quality.
- Controls:
- Use of fire-resistant FRP panels and aluminum frames, certified for offshore use.
- Routine inspection and maintenance schedules.
- Engineering review of habitat and skid loading versus deck constraints.

Logistical and Integration Risks

- Hazard: Delays or errors during habitat/skid installation, movement, or interface with well equipment.
- Consequence: Minor to Moderate—operational delays, potential for unsafe temporary setups.
- Likelihood: Low with careful planning and modular design.
- Controls:
- Prefabrication and modularization for efficient transport and assembly.
- Interface management with wellhead equipment and existing control/monitoring systems.
- Comprehensive pre-job planning and hazard identification (HAZID) workshops.

Human Factors and Training

- Hazard: Operator error due to unfamiliarity with pressurized habitat systems or emergency procedures.
- Consequence: Moderate—may lead to ineffective response in emergencies.
- Likelihood: Low-Moderate, mitigated by training.
- Controls:
- Comprehensive training and competency assessment for operating personnel.
- Clear written procedures for habitat pressurization, shutdown, and evacuation.
- Regular drills and refresher training sessions.

Summary of Mitigations

- Positive-pressure habitat engineering with robust pressure and gas monitoring, automated safety shutdowns, and ATEX/IECEx certification.
- Use of high-integrity materials and modular, prefabricated construction for both habitat and power skid.
- Integrated safety and control systems, including redundant instrumentation and remote telemetry.
- Comprehensive personnel training, maintenance, and continuous improvement through operational feedback.

The Project Results and Impact

The deployment delivered transformative results

- First-of-its-kind Application: This was the first documented use of a temporary power skid housed in a pressurized habitat for ESP operations in Abu Dhabi's offshore fields.
- Early Production Unlock: Wells were brought online months ahead of schedule, securing valuable early production and expediting reservoir evaluation and revenue generation.
- Significant Cost Savings: The implementation avoided the need for expensive and logistically intensive well activation methods, and by deferring permanent infrastructure investments, captured more than \$10 million USD in net gains.

- **Enhanced Well Testing and Commissioning:** The flexibility to test and commission ESPs immediately after installation reduced the risk of deferred operational issues and facilitated informed decision-making for ongoing field development.
- **Operational Precedent:** The success of this approach sets a clear precedent for future offshore ESP projects, highlighting the value of adaptable, temporary infrastructure solutions in dynamic field development environments.

Lessons Learned and Future Outlook

The successful deployment of a temporary power skid with pressurized habitat offers several key lessons for operators facing similar challenges worldwide:

- **Innovation through Adaptation:** Repurposing onshore or standard equipment for offshore use—when appropriately engineered for safety—can unlock value and flexibility previously considered unattainable.
- **Safety without Compromise:** Creative engineering solutions such as pressurized habitats ensure that operational innovation does not come at the expense of safety or regulatory compliance.
- **Collaboration and Planning:** Cross-disciplinary collaboration—bringing together electrical, safety, mechanical, and operations teams—was essential for the seamless integration of the solution within existing offshore workflows.
- **Potential for Broader Application:** The demonstrated benefits suggest that temporary power solutions could be effectively applied in other offshore settings, and potentially for other forms of downhole equipment where infrastructure limitations exist.

Conclusion

The first successful deployment of a temporary power skid within a pressurized habitat for ESP operations in Abu Dhabi's offshore field represents a significant leap forward in offshore artificial lift technology. By addressing the dual challenges of infrastructure limitations and stringent safety requirements, the solution not only delivered immediate operational and financial gains but also laid the foundation for future innovation in offshore field development. As offshore projects continue to demand greater flexibility, efficiency, and cost control, the adoption of temporary, adaptable solutions like this will play an increasingly central role in maximizing value and accelerating production timelines.

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REFERENCE

1. Remote VSD Installation of a Single-Well Platform Available, Paper presented at the SPE Middle East Artificial Lift Conference and Exhibition, Manama, Bahrain, October 2024. Paper Number: SPE-221542-MS

2. The Best-In-Class Implementation of Dual ESP Technology in an Abu Dhabi Mature Offshore Oil Field to Successfully Extend Well Life with Significant Contribution to Sustainability Paper presented at the ADIPEC, Abu Dhabi, UAE, October 2023. Paper Number: SPE-216378-MS
3. Case Study: First Successful Offshore ESP Project in Saudi Arabia Paper presented at the SPE Saudi Arabia Section Technical Symposium, May 9-11, 2009 Paper Number: SPE-126066-MS